

Set Abacus

1. Datapath with control for load and store instruction combined as given in the slides.
2. No. For LW and SD, the ALU control does not use instruction bit 30 or bits 14–12.

Both LW and SD use the ALU only to compute the effective address by adding the base register and the immediate offset. Therefore, the ALU operation is fixed to ADD, and the ALU control is directly determined by the opcode. Instruction bits funct3 (14–12) and funct7 bit 30 are relevant for distinguishing operations in R-type or some I-type arithmetic instructions, not for load/store address calculation.

3. Solve
 - a. RegWrite control signal is on/enabled for R-type, I-type(non-ld), Load instructions only.
So, $15 + 20 + 20 = 55\%$ is the answer.
 - b. Final output comes from ALU only in R-type and I-type(non-ld) instructions only.
So, $15 + 20 = 35\%$ is the answer.
 - c. The processor needs to perform sign extension wherever there is an immediate value.
So, I-type (non-ld), Load, Store, Branch and Jump = $20 + 20 + 10 + 25 + 10 = 85\%$

Set Byte

1. Datapath with control for store instruction as given in the slides.
2. No. For LW and SD, the ALU control does not use instruction bit 30 or bits 14–12.
Both LW and SD use the ALU only to compute the effective address by adding the base register and the immediate offset. Therefore, the ALU operation is fixed to ADD, and the ALU control is directly determined by the opcode. Instruction bits funct3 (14–12) and funct7 bit 30 are relevant for distinguishing operations in R-type or some I-type arithmetic instructions, not for load/store address calculation.
3. Solve
 - a. RegWrite control signal is on/enabled for R-type, I-type(non-ld), Load instructions only.
So, $10 + 25 + 15 = 50\%$ is the answer.
 - b. Final output comes from ALU only in R-type and I-type(non-ld) instructions only.
So, $10 + 25 = 35\%$ is the answer.
 - c. The processor needs to perform sign extension wherever there is an immediate value.
So, I-type (non-ld), Load, Store, Branch and Jump = $25 + 15 + 15 + 20 + 15 = 90\%$